

SPRING 2025 (CSE423 QUIZ)

Time: 30 Minutes

Name: _____ ID: _____

Section: _____

A

1. Convert the following color from **HSV** to **RGB**: 5
 $\text{HSV} = (270^\circ, 40\%, 90\%)$
2. Convert the following color from **CMY** to **HSV**: 5
 $\text{CMY} = (255, 85, 170)$
3. Answer these short questions: 5
 - a. What is a **vanishing** point in projection? (1)
 - b. What is the maximum number of vanishing points that can exist in a projection? (1)
 - c. Which type of projection involves **vanishing** points? (1)
 - d. If you move the projection plane away from the object, will the projected image become larger or smaller? Answer for both **perspective** projection and **parallel** projection. (2)
4. You are an architect working on designing a new skyscraper. The building is being constructed on a large plot of land, and you need to create a visual 3D model that will be used to present the design to your clients. You need to make sure the model looks realistic and that the depth of the building is properly conveyed on a 2D screen.
In order to see how the building looks from a certain vantage point, you create a 3D coordinate system where:
 - The origin (0, 0, 0) is at the base of the building.
 - The **negative** x-axis represents the front of the building.
 - The **positive** z-axis represents the height of the building.You have set your projection plane 100 meters (in direction of negative x-axis) in front of the building (parallel to the YZ plane). Now answer the following question:
5
 - a. Which type of projection will you use? Why? (2)
 - b. If you wanted to get proper measurements instead of a realistic view, would you choose any different type of projection? (1)
 - c. Suppose you have chosen to use perspective projection, and you have placed your camera 50 meters away from the projection plane along the x-axis and at a height of 40 meters above the ground level. What will be the coordinates of the camera? (2)

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B

1. Convert the following color from **HSV** to **RGB**: 5
 $\text{HSV} = (30^\circ, 40\%, 90\%)$
2. Convert the following color from **CMY** to **HSV**: 5
 $\text{CMY} = (85, 170, 255)$
3. Answer these short questions: 5
 - a. What is a **vanishing** point in projection? (1)
 - b. What is the maximum number of vanishing points that can exist in a projection? (1)
 - c. Which type of projection involves **vanishing** points? (1)
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C

1. Convert the following color from **HSV** to **RGB**: 5
 $\text{HSV} = (150^\circ, 40\%, 90\%)$
2. Convert the following color from **CMY** to **HSV**: 5
 $\text{CMY} = (85, 255, 170)$
3. Answer these short questions: 5
 - a. What is a **vanishing** point in projection? (1)
 - b. What is the maximum number of vanishing points that can exist in a projection? (1)
 - c. Which type of projection involves **vanishing** points? (1)
 - d. If you move the projection plane away from the object, will the projected image become larger or smaller? Answer for both **perspective** projection and **parallel** projection. (2)
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